

Consider yourself Senior Test Automation Engineer

Help me to build a scalable, data-driven Selenium automation framework.

[Important] Do not provide the implementation immediately. Instead, act as a Solution Architect and gather my requirements using the Socratic questioning technique.

Rules to be followed for the questioning

[Strict] Ask exactly one question at a time.

[Mandatory] Each question should provide multiple-choice options (A, B, C, D...) along with a brief explanation of each option.

[Mandatory] Wait for my response before asking the next question.

[Strict] Ask only the questions necessary to fully understand my requirements.

[Strict] Do not make assumptions unless I explicitly ask for recommendations.

[Important] Ask only 10 to 12 questions

Once all the requirements are gathered:

[Important] Summarize my chosen requirements.

[Important] Explain the architecture and design decisions.

[Important] Provide the recommended project structure.

[Important] Implement the complete framework step by step with well-commented Java code.

[Important] Explain why each design choice was made and discuss possible alternatives.

[Important] Follow industry best practices for Selenium, TestNG, Maven, Page Object Model, and data-driven testing.

Thank you.

Selected Requirements so far

-  Primary Goal: **UI Functional Automation**
-  Technology Stack: **Java + TestNG**
-  Build Tool: **Maven**
-  Test Data Management: **Multiple Data Sources (Excel, JSON, CSV, Database) through a common abstraction**
-  Browser Support: **All major browsers (Chrome, Firefox, Edge, and Safari where applicable)**

- ☒ WebDriver Management: **ThreadLocal WebDriver** for parallel execution
 - ☒ Design Pattern: **Page Object Model (POM)**
 - ☒ Reporting & Logging: **Support multiple reporting tools (Extent Reports and Allure) through an abstraction, with Log4j2 for logging**
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Question 9 of 10

What level of test execution capability should the framework support?

A. Sequential Execution Only

- Tests run one after another.
- Simple to configure but slower for large test suites.

B. Parallel Execution (Local Machine)

- Execute tests in parallel using TestNG and ThreadLocal WebDriver.
- Significantly reduces execution time on a single machine.

C. Selenium Grid / Docker

- Execute tests across multiple machines, browsers, and operating systems.
- Ideal for scalable enterprise and distributed execution.

D. Cloud Execution (BrowserStack, Sauce Labs, LambdaTest, etc.)

- Run tests on real browsers/devices in the cloud.
- Suitable for cross-browser and cross-platform validation.

E. All of the Above

- Framework should support sequential, local parallel, Selenium Grid/Docker, and cloud execution through configuration without code changes.
- Provides maximum flexibility and scalability.

Please reply with the option (A/B/C/D/E).

E

Thank you.

Selected Requirements so far

- ☒ Primary Goal: **UI Functional Automation**
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 - ☒ WebDriver Management: **ThreadLocal WebDriver for parallel execution**
 - ☒ Design Pattern: **Page Object Model (POM)**
 - ☒ Reporting & Logging: **Multiple reporting support (Extent Reports and Allure) with Log4j2**
 - ☒ Execution Modes: **Sequential, Local Parallel, Selenium Grid/Docker, and Cloud execution (configurable without code changes)**
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Question 10 of 10

Which advanced framework capabilities should be included?

A. Core Framework

- Configuration management
- Explicit waits
- Screenshot capture on failure
- Retry mechanism
- Utility classes
- Environment-specific configuration

B. Enterprise Framework

- Everything in **A**, plus:
- CI/CD integration (e.g., Jenkins, GitHub Actions, Azure DevOps)
- Cross-browser execution
- Automatic driver management
- Test categorization (Smoke, Regression, Sanity)
- Listeners and hooks

C. Advanced Enterprise Framework

- Everything in **B**, plus:
- API utilities for test setup/cleanup

- Database validation utilities
- Email notification after execution
- Custom annotations
- Soft assertions
- Data masking for sensitive test data
- Self-healing locator support (where applicable)
- Custom exception handling
- Framework extensibility using interfaces and factory patterns

D. Other

- Please specify the capabilities you want.

Please reply with the option (A/B/C/D).

C

Thank you. We have now gathered all the required information.

In the next response, I will provide:

1. **A summary of your selected requirements**
2. **A detailed architecture with design decisions**
3. **The recommended Maven project structure**
4. **A step-by-step implementation of the complete framework**
5. **Well-commented Java code for every class**
6. **Explanations for each design decision and possible alternatives**
7. **Industry best practices** for Selenium, TestNG, Maven, Page Object Model, data-driven testing, parallel execution, reporting, and CI/CD integration.

Final Requirements Summary

Category	Selected Requirement
Framework Goal	UI Functional Automation
Programming Language	Java
Test Framework	TestNG

Category	Selected Requirement
Build Tool	Maven
Test Data	Multiple data sources (Excel, JSON, CSV, Database) through a common abstraction
Browser Support	Chrome, Firefox, Edge, Safari
Driver Management	ThreadLocal WebDriver
Design Pattern	Page Object Model (POM)
Reporting	Extent Reports + Allure through an abstraction
Logging	Log4j2
Execution	Sequential, Local Parallel, Selenium Grid/Docker, Cloud (BrowserStack/Sauce Labs/LambdaTest)
Framework Level	Advanced Enterprise Framework (API utilities, DB utilities, email notifications, custom annotations, soft assertions, data masking, self-healing support, custom exceptions, extensible design)

I have all the information needed and will use these requirements as the basis for designing and implementing the framework.